

Questions, and where the data comes from

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MA Computational Social Science, UC3M
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Where we are

1. Why design at all? done
2. **What is the question, and where would the data come from? today**
3. Do the numbers mean what you think they mean?
4. What comparisons can we use to establish causal claims?
5. What variation can you exploit, and how far does the answer travel?

Roadmap

From topic to research question

Theory, mechanisms, levels

Where does the data come from?

Ten things about big data sources

Paper: The Parable of Google Flu

Before next week

What is a research question?

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- **Even though** this 'argument' can be something anecdotal that we later develop into a proper, abstract theoretical argument
 - And it would actually look into something like this:
 - Previous > 'Anecdotal argument' > RQ > (Proper) Theory > Hs

What is a research question? Example

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- 'Main' RQ: Does online media cause an increase in lone-wolf political violence?
 - RQ1: Has political violence increased recently?
 - RQ2: Political violence now more likely by individual young men?
 - RQ3: Is online radicalization the cause of this?

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 - RQ3: Is online radicalization the cause of this?
- Theoretical argument, hypotheses, expectations, etc

Good RQs, in brief

1. Empirically **answerable**

→ i.e. you can answer it with data

2. Theoretically **relevant**

→ i.e. it helps you learn something about your theory/argument

Good RQs, more in detail

1. Consider potential results of the analyses

- if you found X, does that answer the question? causally?
- example: are kids who play videogames often more aggressive?
- does that inform a theory on the aggressiveness effect of VGs?
(you can even try to do a better RQ without causal ID)

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- do you have the data? is it possible to do it? (e.g. re-offenders)
- also: is there any design or strategy to answer it?

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3. Keeping it simple and narrow

- what are the causes of economic underdevelopment? vs. does exposure to natural disasters hinder economic development?

Good RQs, more in detail

This:

- Has political violence increased recently?
- Political violence now more likely by individual young men?
- Is online radicalization the cause of this?

vs this:

- What's the link between online media and political violence?
(too broad)
- Is online media the cause of the increase of political violence among young men? (many questions in one)
- Why does online media drive political violence? (??)

Example on generating RQs

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 - So they already suggest which variation to look at
- Imagine you have the following argument:

How good students do at school depends more on the peers they are surrounded by than on the quality of the teaching they receive
- Which RQ could let us test this?

Stories, RQs, and theories

- There're no exclusive definitions of 'stories' and theories
- It's just about getting to a sufficient level of abstraction
- Often, you start with a story or example, and then you move up the ladder for both theory and RQs until you get to a general theory tested with a RQ

Generating theories

- No recipe for this, everyone generates theories *all the time*
- Usually it refers to an analytical argument that explains something
 - It could also be a descriptive or predictive theory, but even in those cases there's probably an explanation underneath
- Developed inductively, from descriptive data to general explanations
- My advice: if you can't tell a story out of the theory, you're not there yet (i.e. need to be able to travel from/to abstraction)

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- **Q:** How to identify a **good theory**?

Evaluating theories

1. Simple

Evaluating theories

1. Simple
2. Internally coherent and able to explain variation

Evaluating theories

1. Simple
2. Internally coherent and able to explain variation
3. Testable

Example (of the whole process)

Internacionales UMU
@internacionalUMU

¿Y sabías que encuentran **#trabajo** más rápido que los estudiantes que no realizan **#movilidad**?

3 de cada 4 graduados Erasmus+ consideran que su **#experiencia** en el extranjero es beneficiosa para encontrar su primer trabajo, y el 80% encontraron trabajo 3 meses después de graduarse.

Translate Tweet



The advertisement graphic features a blue background with a hand pointing at a glowing orange figure icon. To the right, there are two smaller blue circles, each containing a white figure icon. The text is in white and yellow. The top left corner has the Erasmus+ logo and the text 'Erasmus+ ESTUDIOS'. The main headline is 'DIFERENCIATE DEL RESTO'. Below it, a paragraph states: 'Los estudiantes Erasmus+ encuentran trabajo más rápido que los estudiantes que no realizan movilidad.' At the bottom left, vertical text reads 'ELIGE TU DESTINO ELIGE ERASMUS'. At the bottom center, it says 'Solicitud abierta hasta el 14/12/20' and 'erasmus.um.es'. At the bottom right, there is a red box with 'UNIVERSIDAD DE MURCIA' and a small circular logo.

ELIGE TU DESTINO
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UNIVERSIDAD DE
MURCIA

12:05 PM · Nov 23, 2020 · Twitter Web App

Example

- That's some descriptive evidence that could inspire an anecdote
- The **anecdotal argument** (think of a story)
- The **research question**
- The 'proper' **theory**
- The **hypotheses?**

Example

- That's some descriptive evidence that could inspire an anecdote
- The **anecdotal argument** (think of a story)
 - My friend John who went on Erasmus has more money than my other friend who couldn't go, and also, John managed to get a job because his father is partner at a local firm
- The **research question**
 - Is there a causal effect of Erasmus on labor market early success? Is the effect mediated by household income?
- The 'proper' **theory**
 - Going on Erasmus does not have any causal effect on getting a first job, the relationship is explained by the confounding effect of income
 - Or: Positive effect among high-income students because they have access to informal networks where this experience is valued
- The **hypotheses?**

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 - Which essentially means being able to say what you expect to see given a theory
And ideally, knowing what you need to see in order to discredit the theory
- Imagine that I have the theory that my knee hurts when I do sports on cold days
 - Simplifying it, we have a 2x2 situation:

Hypotheses: what would you expect to observe?

	<i>Cold day</i>	<i>Hot day</i>
<i>Run</i>		
<i>Didn't run</i>		

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	<i>Cold day</i>	<i>Hot day</i>
<i>Run</i>	Pain	Not pain
<i>Didn't run</i>	Not pain	Not pain

What if you observe this? **New theory? Test?**

	<i>Cold day</i>	<i>Hot day</i>
<i>Run</i>	Pain	Not pain
<i>Didn't run</i>	Pain	Pain

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Computational methods and theory

- Limitations of *data mining*
- Focus on the what rather than on the why
- Problems with machine learning
 - Example of predicting ice cream sales

Why bother about theory?

CIVIL UNREST

Predicting Civil Conflict: What Machine Learning Can Tell Us

Computer programs can be used as early warning systems, allowing the global community to act before violence erupts.

Mechanisms briefly

- A **mechanism** is basically the **how** (or why) of a relationship
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Mechanisms briefly

- A **mechanism** is basically the **how** (or why) of a relationship
- e.g., we know that the flu gives us fever
 - flu > fever
- What's the mechanism?
 - flu > *immune system detects infection* > Δ *body temp* > fever
- Good thing about mechanisms is that we can try to test them

Testing mechanisms ($\approx$ sub-research questions)

- Let's go back to the Erasmus example
- If our theory is that the effect of going on Erasmus is higher for high-income students due to their access to social networks,
what's the mechanism?
- And how could we try to **test it**?
 - (Think about the sub-RQs)

Complexity of the social world and micro/macro



- Why do mass protests emerge?

Levels of explanation

Can you think of...?

- Macro-level mechanisms
- Micro-level mechanisms

Levels of explanation

Can you think of...?

- Macro-level mechanisms
- Micro-level mechanisms
- What's the point of macro-level explanations, actually?

Recap

- Questions on theory, RQ, or mechanisms?

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Last week, in one line

Designed

answers a question **you** had

Found

answers a question **someone else** had.
Or no question at all

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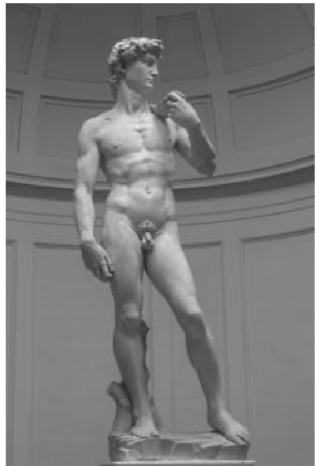
Found

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Today: what follows from being on the right-hand side.



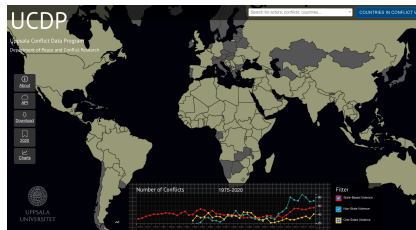
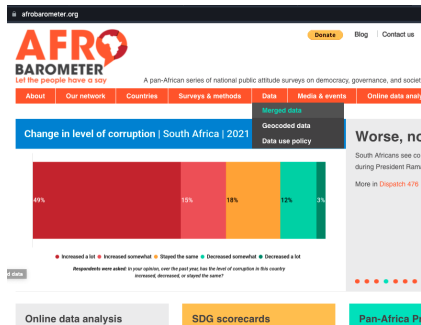
Readymade



Custommade

Figure 1.2: *Fountain* by Marcel Duchamp and *David* by Michelangelo. *Fountain* is an example of a readymade, where an artist sees something that already exists in the world and then creatively repurposes it for art. *David* is an example of art that was intentionally created; it is a custommade. Social research in the digital age will involve both readymades and custommades. Photograph of *Fountain* by Alfred Stieglitz, 1917 (Source: *The Blind Man*, no.

Designed data: someone decided what to record



A survey, and a hand-coded conflict dataset. Someone wrote the questions, or the coding rules.

Found data: nobody decided anything, for you

[nature](#) > [nature communications](#) > [articles](#) > [article](#)

Article | [Open Access](#) | [Published: 20 August 2018](#)

Sequences of purchases in credit card data reveal lifestyles in urban populations

[Riccardo Di Clemente](#), [Miguel Luengo-Oroz](#), [Matias Trivizano](#), [Sharon Xu](#), [Bapu Vaitla](#) & [Marta C. González](#) 

[Nature Communications](#) **9**, Article number: 3330 (2018) | [Cite this article](#)

14k Accesses | 31 Citations | 268 Altmetric | [Metrics](#)

 This article has been [updated](#)

Abstract

Zipf-like distributions characterize a wide set of phenomena in physics, biology, economics, and social sciences. In human activities, Zipf's law describes, for example, the frequency of appearance of words in a text or the purchase types in shopping patterns. In the latter, the uneven distribution of transaction types is bound with the temporal sequences of purchases of individual choices. In this work, we define a framework using a text compression technique

RESEARCH ARTICLE

Faces in the crowd: Twitter as alternative to protest surveys

[Christopher Barrie](#) ^{1,*,} [Arun Frey](#) ^{2,3}

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Abstract

Who goes to protests? To answer this question, existing research has relied either on retrospective surveys of populations or in-protest surveys of participants. Both techniques are prohibitively costly and face logistical and methodological constraints. In this article, we investigate the possibility of surveying protests using Twitter. We propose two techniques for sampling protestors on the ground from digital traces and estimate the demographic and ideological composition of ten protestor crowds using multidimensional scaling and machine-learning techniques. We test the accuracy of our estimates by comparing to two in-protest surveys from the 2017 Women's March in Washington, D.C. Results show that our Twitter sampling techniques are superior to hashtag sampling alone. They also approximate the ideology and gender distributions derived from on-the-ground surveys, albeit with some bias, but fail to retrieve accurate age group estimates. We conclude that online samples are yet unable to provide reliable representative samples of offline protest.

Credit card transactions and social media posts, repurposed as research data.

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- With **found** data, you have to **reconstruct** it, from the purpose of whoever built the system
- So: who made this, and what were they trying to do?

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Before next week

Salganik's ten characteristics

Why we want them

Big

Always-on

Non-reactive

Why designs break

Incomplete

Inaccessible

Non-representative

Drifting

Algorithmically confounded

Dirty

Sensitive

Salganik, *Bit by Bit*, ch. 2. Free online.

Big, always-on, non-reactive

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 - no social desirability bias. But no informed consent either

Non-representative

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- So: is a non-representative sample useless?

The Xbox poll

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Wang, Rothschild, Goel & Gelman (2015), *Int. J. Forecasting* 31(3).

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- **The platform changes**

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A measure can stop meaning what it meant,
without anything looking wrong .

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Lazer, Kennedy, King & Vespignani (2014), *Science* 343(6176). This week's paper.

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 - people searched differently because the *product* changed

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- The platform is in your causal diagram (week 4)

Inaccessible, and the post-API age

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Freelon (2018), Computational Research in the Post-API Age, *Political Communication* 35(4).

Inaccessible, and the post-API age

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- A design that depends on access you do not control is a design that can **stop working**
- And nobody can check your results afterwards

Freelon (2018), Computational Research in the Post-API Age, *Political Communication* 35(4).

Dirty, and sensitive

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- **Sensitive:** much of it is about identifiable people who never agreed to anything
 - “it’s public” is not the same as “they consented”
 - aggregation and linkage can re-identify

(week 5)

Roadmap

From topic to research question

Theory, mechanisms, levels

Where does the data come from?

Ten things about big data sources

Paper: The Parable of Google Flu

Before next week

Why this paper

- Three pages, no equations
- Measurement validity, drift, and algorithmic confounding **in one case**
- And a well-resourced team getting it wrong in public

In pairs: four minutes each

1. What was GFT actually **measuring**?
2. Whose behaviour changed: the public's, or Google's?
3. Would **more data** have fixed it?
4. What would you have had to do in 2009 to see this coming?

Groups of four: redesign it

You have Google's search logs and CDC data, in 2009.

Build something that does not break the same way.

Eight minutes. One change, and what it buys you.

The design lesson

Your instrument was built by someone else,
for their own purposes,
and they will keep changing it.

Which is a measurement problem. That is next week.

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Before next week

- **Read:** Müller-Crepon, Hunziker & Cederman, *Roads to Rule, Roads to Rebel*
 - Longer than Google Flu. Skip the statistics, read the measurement

Before next week

- **Read:** Müller-Crepon, Hunziker & Cederman, *Roads to Rule, Roads to Rebel*
 - Longer than Google Flu. Skip the statistics, read the measurement
- **Memo 2**, in pairs, Aula Global by **Monday night**:
 - Take a source of found data you could realistically get hold of
 - Who produced it, and why? Who or what is missing from it?
 - Name one question it could answer well, and one it could not
 - Half a page

Questions?